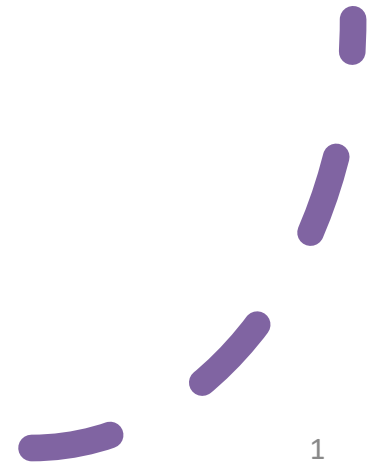


Topic 3 – Week 3

Data Flow Diagrams (DFD)



Learning Objectives

At the end of this lesson, students should be able to:

- Explain what a DFD is and why it is used
- Identify the four core DFD components
- Distinguish a context diagram from a Level 0 DFD
- Describe how data moves through a system
- Understand balancing and decomposition
- Avoid common DFD mistakes

Why Diagrams Matter in Analysis

Diagrams turn long verbal explanation into a visual form that is easier to discuss and review.

- They help teams see structure, flow, boundaries, and relationships more clearly.
- A good diagram does not replace thinking, but it makes thinking easier to share.
- Key words: visual model, communication, structure, clarity.

Choosing the Right Diagram

- Different diagrams answer different questions. Some show process flow, some show data structure, and some show interactions.
- Analysts should choose a diagram based on what they want to explain, not simply use every possible diagram.
- A useful diagram is one that helps decision making and understanding.

Example: use ERD for data structure and BPMN for business workflow.

What Is a Data Flow Diagram?

- A Data Flow Diagram, or DFD, shows how data moves through a system and how processes transform that data.
- It focuses on what data comes in, what processing happens, where data is stored, and what goes out.
- A DFD does not usually show timing or detailed control logic.

A DFD answers these questions:

- What data comes into the system?
- What processing happens to that data?
- Where is the data stored?
- What information goes out of the system?

What a DFD does not show

A DFD does not mainly show:

- Screen interface design
- Program code
- Database table structure in detail
- Timing sequence like a flowchart
- Detailed business responsibility like BPMN

Quick contrast

Use case diagrams answer the question:

****Who wants what from the system?****

DFDs answer the question:

****How does data move through the system?****

Why both are useful

- Use cases show user goals and interactions
- DFDs show data input, processing, storage, and output

The four basic DFD elements

The four basic elements are: process, data flow, data store, and external entity.

1. Process

- A process changes input data into output data.

Examples:

- Record Request
- Verify Documents
- Update Request Status
- Generate Notification

2. Data Flow

- A data flow is the movement of data from one part of the system to another.

Examples:

- Request Details
- Student Information
- Approval Decision
- Notification Message

The four basic DFD elements

3. Data Store

- A data store is where data is kept for later use.

Examples:

- Student Records
- Request Database
- Service Type List

4. External Entity

- An external entity is outside the system but sends or receives data.

Examples:

- Student
- Staff
- Payment Gateway
- Notification Service

Rules for naming DFD elements

Element	Process names	Data Flow	Data Store	External Entity
Memory trick	Use a verb phrase	Noun phrase	Noun	Role or organization name
Examples	• Review Request • Update Request Status	• Request Details • Student Information	• Student Records • Request Records	• Student • Academic Affairs Staff

Good vs Bad DFD names

1. Good names

- Process: **Update Request Status**
- Data flow: **Approval Decision**
- Data store: **Status History**
- External entity: **Student**

2. Weak names

- Process: **Request**
- Data flow: **Data**
- Data store: **File**
- External entity: **User 1**

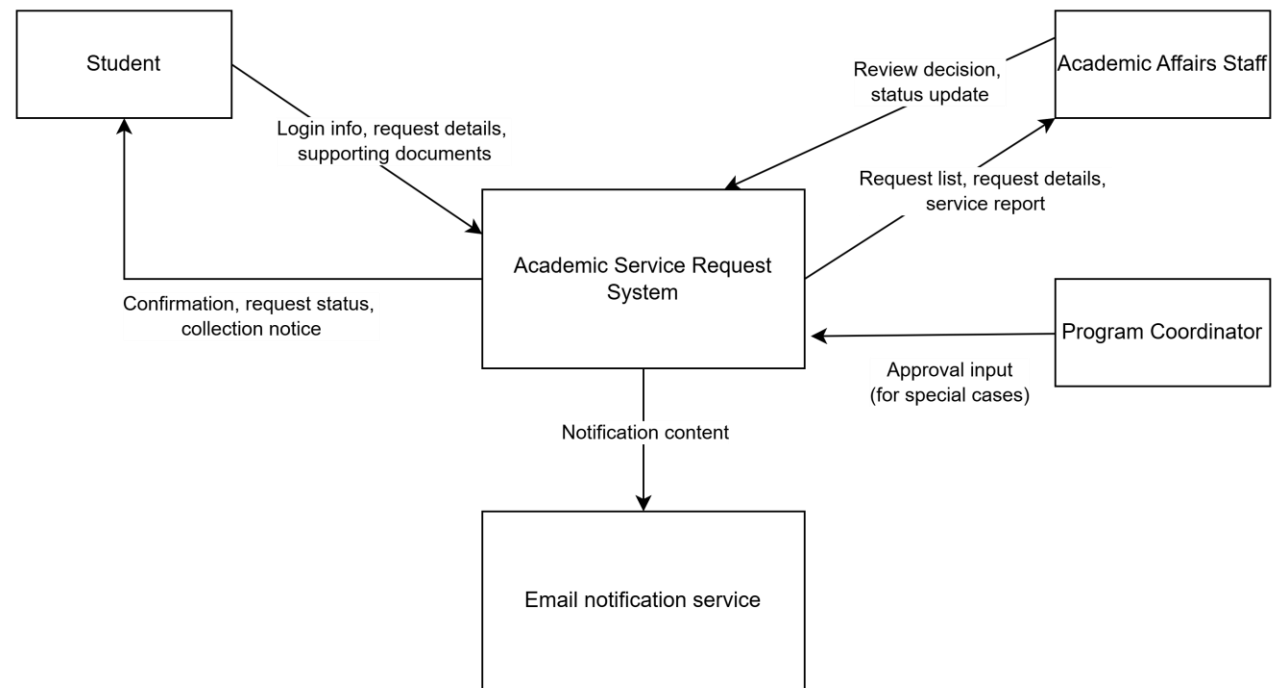
Key rule

The name should make the element meaningful without extra explanation.

Context Diagram

A context diagram is the highest-level DFD. It shows the entire system as one process and only shows external entities and major data flows.

Context diagram for
The Student Academic Service Request System



Level 0 DFD

A Level 0 DFD decomposes the whole system into several major processes while also showing data stores and major data flows.

Example: Level 0 DFD for Student Academic Service Request System

Main processes

- 1.0 Receive and Record Request
- 2.0 Review and Verify Request
- 3.0 Update Request Status
- 4.0 Notify Student
- 5.0 Generate Report

Data stores:

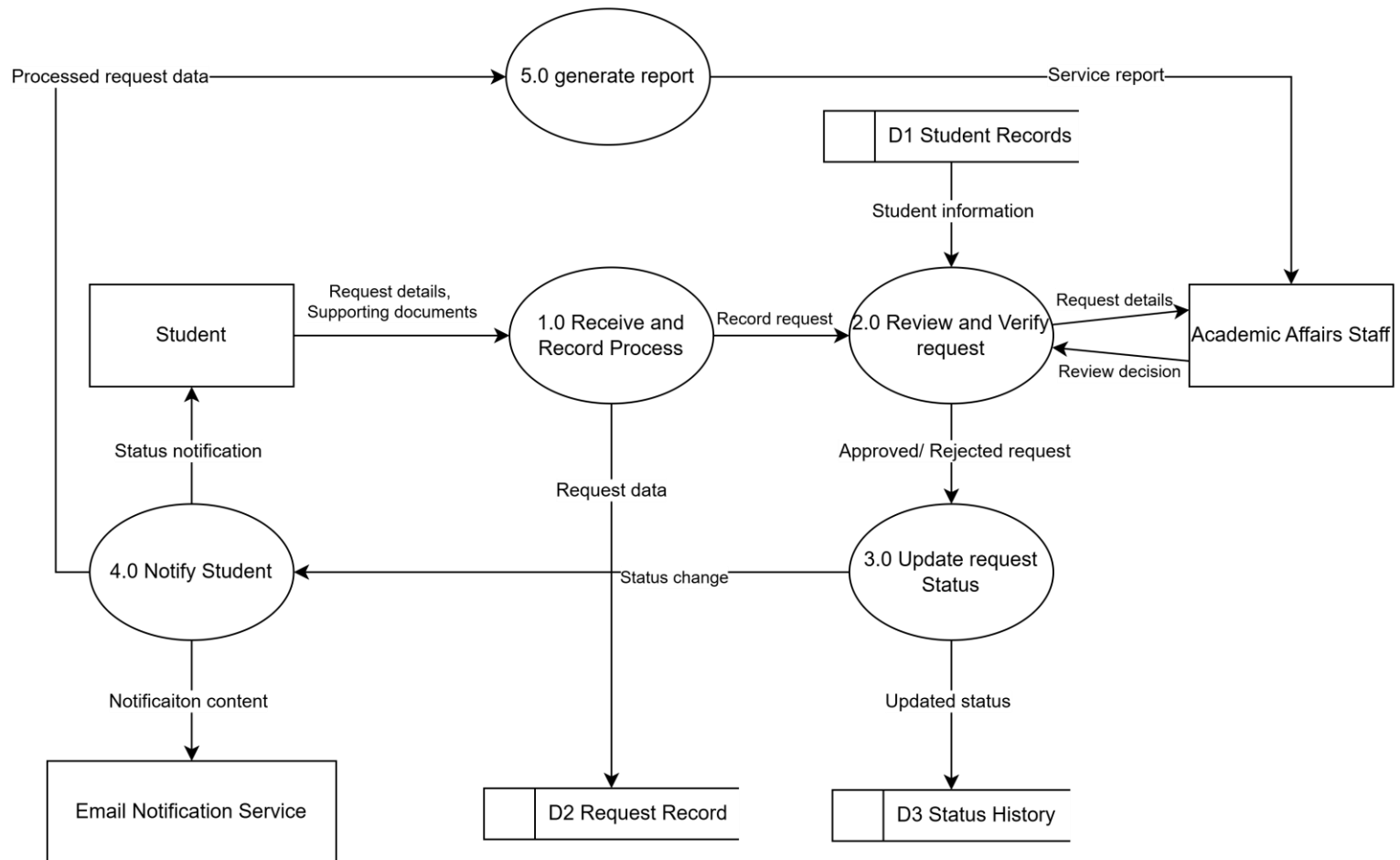
- D1 Student Records
- D2 Request Records
- D3 Status History

Level 0 DFD (Logical flows)

Example logical flows

- Student sends **Request Details** to Process 1.0
- Process 1.0 stores **Request Data** in D2 Request Records
- Process 2.0 reads **Student Information** from D1
- Staff sends **Review Decision** to Process 2.0
- Process 3.0 writes **Updated Status** to D3 Status History
- Process 4.0 sends **Notification Content** to Email Notification Service
- Process 5.0 sends **Service Report** to Staff

Level 0 DFD for the Student Academic Service Request System



Decomposition and leveling

Leveling means breaking a high-level process into lower-level subprocesses when more detail is needed. The DFD emphasizes that decomposition should be logical and only go deeper when necessary.

Example

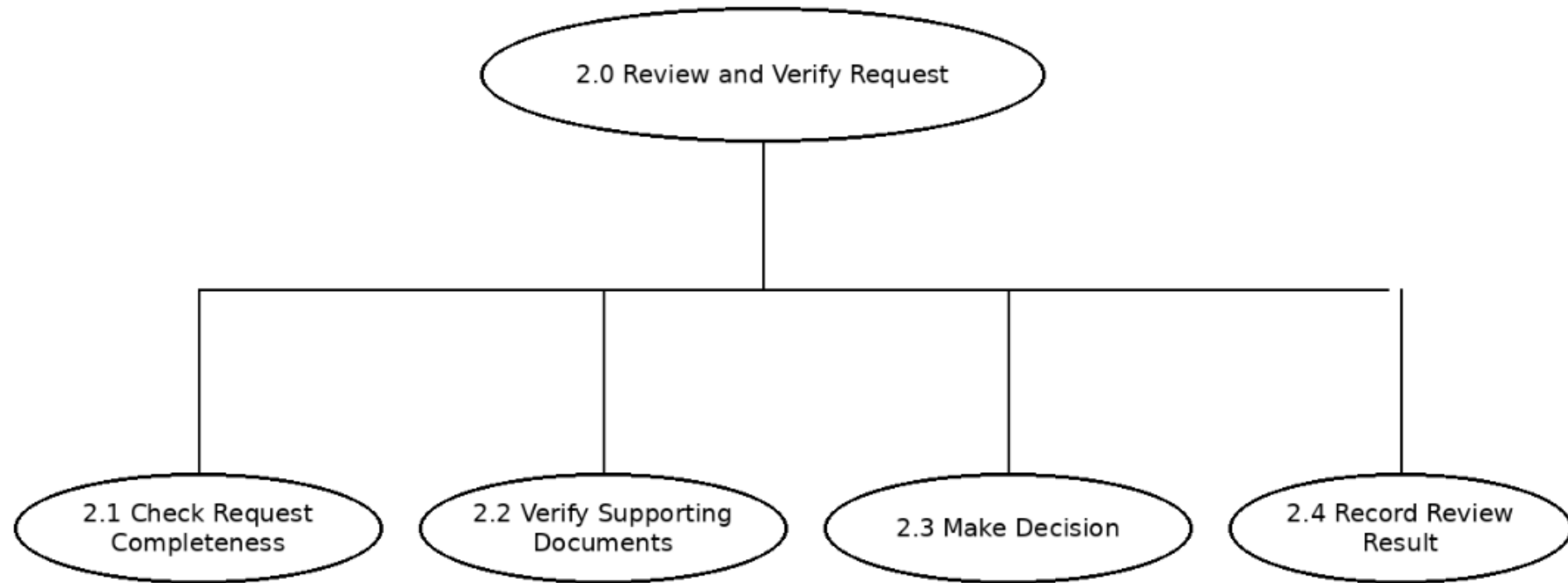
Process **2.0 Review and Verify Request** may be decomposed into:

- 2.1 Check Request Completeness
- 2.2 Verify Supporting Documents
- 2.3 Make Decision
- 2.4 Record Review Result

Key idea

A lower-level DFD should explain the logic of the higher-level process clearly.

Decomposition



Decomposition = zoom in on one high-level process

Balancing in DFD

Balancing means the major inputs and outputs of a parent process should still appear when that process is decomposed into a lower-level diagram. This protects logical consistency across levels.

Example:

If Process 2.0 receives “Request Details” and outputs “Review Decision,” then the lower-level version of Process 2.0 should still account for those flows.

Balancing in DFD

Parent Process

Request Details

2.0 Review and Verify Request

Review Decision

Lower-level DFD must keep the same major input/output

✓ Request Details

2.1 Check Request
Completeness

2.2 Verify Documents

2.3 Make Decision

2.4 Record Result

✓ Review Decision

Common DFD mistakes

The DFD explicitly mentions common mistakes such as unnamed flows, processes with no input or no output, and invalid direct connections.

Important mistakes

1. Unnamed data flows

Every data flow should have a meaningful label.

2. Process with no input

This is called a “miracle.” Data appears from nowhere.

3. Process with no output

This is called a “black hole.” Data goes in but nothing comes out.

4. Process with insufficient transformation

This is called a “gray hole.” The output is not logically supported by the input.

5. External entity directly connected to data store

In a basic logical DFD, data should normally pass through a process.

6. Data store directly connected to another data store

This is usually not allowed in a basic DFD.

7. Mixing physical and logical details

Do not write things like “click submit button” or “Excel file on desktop” in an early logical DFD.

Use case vs DFD

Use Case Diagram

- Focus: user goals
- Main elements: actor, use case, system boundary
- Typical question: *What does the user want the system to do?*

DFD

- Focus: data movement and transformation
- Main elements: process, data flow, data store, external entity
- Typical question: *How does data move through the system?*

Summary line

Use case focuses on user goals.

DFD focuses on data movement.

Mini example 1:

Online Food Ordering System

External entities

- Customer
- Restaurant
- Payment Gateway

Main processes

- Receive Order
- Process Payment
- Confirm Order
- Update Order Status

Data stores

- Order Records
- Payment Records

Example data flows

- Order Details
- Payment Information
- Payment Confirmation
- Order Status

Mini example 2:

Library Borrowing System

External entities

- Student
- Librarian

Main processes

- Search Book
- Reserve Book
- Borrow Book
- Return Book

Data stores

- Book Records
- Loan Records
- Member Records

Example data flows

- Search Query
- Reservation Request
- Loan Details

Summary – Data Flow Diagrams (DFD)

At the end of this lesson, students should remember:

- DFD shows **data movement and processing**
- 4 elements: **process, data flow, data store, external entity**
- Context Diagram = **whole system view**
- Level 0 DFD = **major internal processes**
- Good DFDs are **clear, balanced, and logically labeled**
- Use case = **user goals**; DFD = **data movement**